SURFACE MOUNT

RF Transformer

TC4-1T+

50Ω 0.5 to 300 MHz

TC4-1TX+ Upgraded Version*

***Addition of Top Hat® Feature Benefits**
- Allows Faster Pick-and-Place
- Enables Visual Identification Marking

FEATURES
- Wide Band 0.5 to 300 MHz, Usable Over 0.2 to 450 MHz
- Good Return Loss
- Impedance Matching
- Plastic Base With Leads
- Aqueous Washable

Generic photo used for illustration purposes only

CASE STYLE: AT224-1A

+RoHS Compliant
The +Suffix identifies RoHS Compliance.
See our website for methodologies and qualifications

APPLICATIONS
- CATV

ELECTRICAL SPECIFICATIONS AT +25°C

Parameter	Frequency (MHz)	Min.	Typ.	Max.	Unit
Impedance Ratio (Secondary/Primary)			4		
Frequency Range		0.5		300	MHz
Insertion Loss[1]	0.5-300 1.5-100			2.0 1.0	dB
Phase Unbalance	0.5-300 1.5-100		4 1		Degree

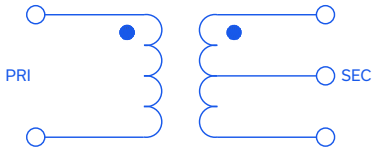
1. Insertion Loss is referenced to mid-band loss, 0.3 dB typ.

ABSOLUTE MAXIMUM RATINGS

Parameter	Ratings
Operating Temperature	-20°C to +85°C
Storage Temperature	-55°C to +100°C
RF Power	0.25 W
DC Current	30 mA

Permanent damage may occur if any of these limits are exceeded.

CONFIG. A



REV. G
ECO-027946
TC4-1T+
MCL NY
260212

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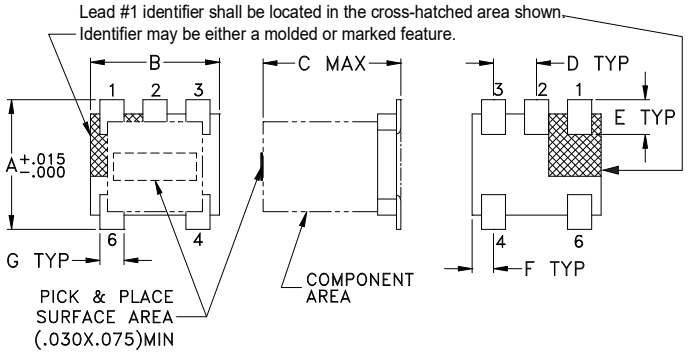
PIN CONNECTIONS

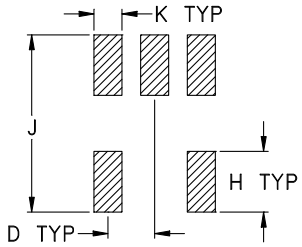
PRIMARY DOT	6
PRIMARY	4
SECONDARY DOT	1
SECONDARY	3
SECONDARY CT	2

OUTLINE DRAWING



PRODUCT MARKING: NA

PCB Land Pattern



Suggested Layout,
Tolerance to be within±.002 in

DEMO BOARD MCL P/N: TB-TC4-1T+

OUTLINE DIMENSIONS (Inch / mm)

A	B	C	D	E	F
.150	.150	.160	.050	.040	.025
3.81	3.81	4.06	1.27	1.02	0.64

G	H	J	K		wt
.028	.065	.190	.030		grams
0.71	1.65	4.83	0.76		0.15

TAPE & REEL INFORMATION: F17

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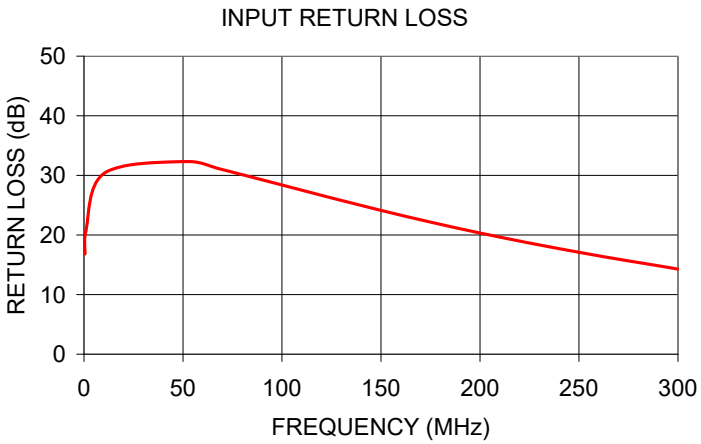
TYPICAL PERFORMANCE DATA

FREQUENCY (MHz)	INSERTION LOSS (dB)	INPUT R. LOSS (dB)
0.5	0.51	16.81
1.0	0.44	20.73
10.0	0.27	30.31
50.0	0.30	32.32
70.0	0.32	30.95
100.0	0.36	28.39
150.0	0.40	24.12
200.0	0.45	20.32
250.0	0.54	17.09
300.0	0.62	14.29



NOTES

A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.

B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.

C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the standard terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/terms/viewterm.html